

Homework 5 - KNN

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1 Finding Gold!

- Do this manually (no code) and show your work. You can use the following table. If there are ties, indicate so and use random selection to break them.

Big Jim found a gold mine and he tries to predict whether or not there is gold at his chosen spot, 200m from his tent (T).

He has made multiple probes at different distances (in meters) from his tent with the following results. Gold (G) or No Gold (N).

0	100	120	140	160	180	200	210	230	250	290
Tent	G	G	N	N	G	?	N	G	N	G

- Use the KNN method with $K = 3$ to make a prediction at Jim's chosen spot.
- Use the KNN method with $K = 5$ to make a prediction at Jim's chosen spot.

	0	100	120	140	160	180	200	210	230	250	290	$\hat{Y}$
	T	G	G	N	N	G	?	N	G	N	G	
distance												
$k = 3 \Rightarrow 2G \text{ vs. } 1N : \hat{Y} = G$						G		N	G			G
$k = 5 \Rightarrow 2G \text{ vs. } 3N : \hat{Y} = N$					N	G		N	G	N		N

a) At 200 meters from the Tent, considering $K = 3$, the prediction at the spot is for "Gold".

b) At 200 meter from the Tent, considering $K = 5$, the prediction at the spot is for "No Gold".